METRIC

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22 August 1988

DETAIL SPECIFICATION

ZIRCONIUM CARBIDE, POWDERED, TECHNICAL

Reactivated after 9 July 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of one type of low hafnium content powdered zirconium carbide hereinafter referred to as the material.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army CCDC – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-650	-	Explosive: Sampling, Inspection and Testing

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E300	-	Standard Practice for Sampling Industrial Chemicals.
ASTM B329	-	Standard Test Method for Apparent Density of Metal Powders and Compounds Using the Scott Volumeter
ASTM B330	-	Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability.

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the material is required and shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing material that complies with the technical requirements of the contract (see 4.5).

3.2 Chemical and physical properties. The chemical and physical properties of the material shall be in accordance with Table I.

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TABLE I. Chemical and physical properties.

Characteristic	Minimum	Maximum
Composition		
Zirconium (Zr plus Hf), weight %	87.4	88.5.
Hafnium (Hf), weight %	---	2.0
Total carbon, weight %	10.8	11.9
Free carbon, weight %	---	1.5
Iron, weight %	---	0.05
Oxygen, weight %	---	0.6
Nitrogen, weight %	---	0.8
Properties		
Apparent density, g/cm^3	1.46	1.83
Average particle diameter, µm	1.5	3.0
Volatiles content, weight %	---	0.10

3.3 DELETED.

3.3.1 DELETED.

3.4 Workmanship. The material shall be uniform, free from contamination, foreign material, or any other defect that would prevent its use for the purpose intended. The material shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. VERIFICATION

4.1 DELETED.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18-35˚C (65-95˚F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95% relative, maximum

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.5)
b. Quality conformance inspection (see 4.6)

4.4 DELETED.

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4.5 Preproduction inspection. Unless otherwise specified in the contract (see 6.2), a preproduction sample consisting of sufficient material to perform the tests of 4.6 shall be manufactured in accordance with 3.1 and forwarded to an activity designated by the procuring activity (see 6.2) for preproduction inspection. The preproduction inspection shall consist of tests and examinations specified in Table II. Acceptance of the preproduction sample shall be based on no defects in the sample. Any production of the material by the contractor, prior to approval of the preproduction sample, shall be at the contractor's risk.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of the tests and examinations specified in Table II on samples selected in accordance with 4.6.2. If any sample does not conform to any requirement of this specification, the lot shall be rejected.

4.6.1 Lot size. A lot shall consist of all material, produced under essentially the same conditions, by the same process, by one manufacturer, at one plant, and to be offered for acceptance at one time.

4.6.2 Quality conformance inspection sampling. Four containers selected at random from each lot shall be individually sampled. If the lot consists of less than four containers, each container shall be sampled. If the lot consists of only one container, the container shall be sampled at two locations.

TABLE II. Preproduction inspection and quality conformance inspection.

Tests and inspections	Requirements	Preproduction inspection	Quality conformance inspection	Test methods
Zirconium	Table I	X	X	4.7.1
Hafnium	Table I	X	X	4.7.2
Total carbon	Table I	X	X	4.7.3
Free carbon	Table I	X	X	4.7.4
Iron	Table I	X	X	4.7.5
Oxygen	Table I	X	X	4.7.6
Nitrogen	Table I	X	X	4.7.7
Density	Table I	X	X	4.7.8
Average particle diameter	Table I	X	X	4.7.9
Volatiles content	Table I	X	X	4.7.10
Workmanship	3.4	X	X	Visual

4.7 Tests.

4.7.1 Zirconium. The zirconium content shall be determined by difference (i.e. sum of the measure elements except hafnium subtracted from the total).

4.7.2 Hafnium. The hafnium content shall be determined by x-ray fluorescence or equivalent. Suitable conditions and sample preparations for x-ray fluorescence are as follows.

a. Conditions.

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Analytical line	Hf 1 L81
Background	41.00
Crystal	LiF 200
Medium	Vacuum
Count time	40 seconds
Anode material	Cr
KV	50
Ma	30
Detector	Scintillation
Collimator	Fine
Vessel	Steel and carbon disk

b. Sample preparation.

Borax beads

(1) containing 5 g $Li_2B_4O_7$ + 0.2 g sample

(2) containing 5 g $Li_2B_4O_7$ + 0.2 g sample + known amounts of pure HfO_2 (1%, 2%, 5%).

4.7.3 Total carbon. Determine total carbon content by the micro-Pregl combustion technique or equivalent.

4.7.4 Free carbon. The free carbon shall be determined as follows:

a. Weigh approximately 1 to 2 g of sample and accurately record weight to 0.0001 g, and transfer to a platinum crucible. Add 25 mL concentrated hydrofluoric acid and then heat on a sand bath. Nitric acid is added dropwise until sample is dissolved. Distilled water is added and the contents of the crucible are quantitatively rinsed into a porcelain filter crucible. Filter crucible and contents are then dried in an oven at 105˚C.

b. The filtration residue is carefully scraped off with a sharp spatula onto a porcelain combustion boat and the carbon content determined by the micro-Pregl combustion technique or equivalent.

4.7.5 Iron.

4.7.5.1 Apparatus.

a. Platinum beakers with matching covers, 200-mL capacity

b. pH meter — hydrogen ion meter, fitted with a glass indicator electrode and a calomel reference electrode, capable of measuring 0.01 pH unit.

4.7.5.2 Reagents.

a. Hydroxylamine hydrochloride solution (100 g/L) – dissolve 50 g of hydroxylamine hydrochloride ($NH_2OH•HCl$) in water and dilute to 500 mL. Store in a refrigerator.

b. Iron standard solution (1 mL = 0.10 mg Fe) – dissolve 0.1000 g of iron wire in 100 mL of HCl (1+1) and dilute with water to 1 L in a volumetric flask.

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c. Ortho-phenanthroline solution (2 g/L) – dissolve 1.0 g of o-phenanthroline monohydrate in hot water. Cool and dilute to 500 mL. Store in a refrigerator.
d. Tartaric acid solution (500 g/L) – Dissolve 500 g of tartaric acid in water and dilute to 1 L.

4.7.5.3 Preparation of calibration curve.

4.7.5.3.1 Calibration solutions. Transfer 0.5, 1.0, 2.0, 3.0, 4.0, 5.0 mL of iron solution (1 mL = 0.10 mg Fe) to six 100-mL beakers. Dilute to 15 mL and proceed in accordance with 4.7.5.3.3.

4.7.5.3.2 Reference solution. Transfer 15 mL of water to a 100-mL beaker and proceed in accordance with 4.7.5.3.3.

4.7.5.3.3 Color development. Add 5 mL of tartaric acid solution, 1 mL of hydroxylamine hydrochloride solution, and 5 mL of o-phenanthroline solution, mixing the solution after each addition. Adjust the pH to 5.5 to 6.0 with solution of NH_4OH (1+3) or HCl (1+3) (sulfuric acid (H_2SO_4) (1+3) may be substituted). Transfer to a 100-mL volumetric flask and dilute to the mark with water. Mix well and place the flask in a water bath at 60 to 70˚C for thirty minutes. Remove from the water bath and cool to room temperature.

4.7.5.3.4 Photometry. Transfer a suitable portion of the reference solution to an absorption cell and adjust the photometer to the initial setting, using a light band centered at 508 nm. While maintaining this photometric adjustment, take the photometric readings of the calibration solutions.

4.7.5.3.5 Calibration curve. Plot the photometric readings of the calibration solutions against milligrams of iron per 100 mL of solution.

4.7.5.4 Procedure.

4.7.5.4.1 Test solution. Transfer 1.000 g of the sample to a platinum beaker, add 10 to 15 mL of water, add 2 mL of HF dropwise in 0.5-mL portions, and cover with the platinum lid. Keep the solution covered except when adding reagents. When most of the sample is dissolved, add 20 mL of hydrofluoric acid. When the initial reaction subsides, heat the solution gently until dissolution is complete, rinse and remove the platinum cover, and boil gently to dense white fumes. Cool and cautiously add 50 to 75 mL of water. Heat gently and swirl, if necessary, to dissolve the salts present. Transfer the clear, cool solution to a 100-mL volumetric flask and dilute to the mark with water.

4.7.5.4.2 Reference solution. Carry a reagent blank through the entire procedure 4.7.5.4.1, using the same amount of all reagents, for use as a reference solution.

4.7.5.5 Color development. Select a suitable aliquot of the solution in accordance with the following table.

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TABLE II. Color development aliquots.

Iron, ppm	Aliquot volume, mL
250 to 500	50
500 to 2000	25
2000 to 3000	10

Transfer the aliquot to a 100-mL beaker, and adjust to a volume of 15 mL if necessary. Proceed as directed in 4.7.5.3.3.

4.7.5.6 Photometry. Take the photometric reading of the test solution as described in 4.7.5.3.4.

4.7.5.7 Calculation. Convert the photometric reading of the sample solution to milligrams of iron by means of the calibration curve. Calculate the parts per million of iron as follows:

$$\text{Iron, ppm} = \frac{A}{B} \text{ x } 1000$$

Where: A = milligrams of iron found
B = grams of sample represented in the aliquot used

4.7.6 Oxygen. The oxygen content shall be determined by the micro Unterzaucher combustion technique or equivalent.

4.7.7 Nitrogen. The nitrogen content shall be determined in accordance with the Dumas method or equivalent.

4.7.8 Density. The density shall be determined in accordance with ASTM B329.

4.7.9 Average particle diameter. The average particle diameter shall be determined in accordance with ASTM B330 using a 6.73 g sample accurately weighed to ± 0.01 g.

4.7.10 Volatiles content. The volatiles shall be determined in accordance with MIL-STD-650, Method 101.5 Moisture and Volatiles (Oven Method).

4.8 Workmanship. Take approximately 200 to 300 grams of the material being inspected and spread out on a clean sheet of white paper. The sample shall be examined with unaided eyes by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter or other defects that would render the material unsuitable for the intended use. Failure to meet the workmanship inspection and/or visual inspection of workmanship during the test methods of 4.5 shall result in lot rejection.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Zirconium carbide powder is intended to be used as a propellant ingredient for the Mark 56 Mod 2 Dual Thrust Rocket Motor and the XM1128.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Material, type, grade, and class required (see 1.2).
c. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Whether a first article sample is required (see 3.1.1).
e. Packaging requirements (see 5.1).

6.3 Material safety data sheets. Contracting officers should identify those activities requiring copies of the completed safety data sheets (SDS) prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313, and 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must identify the activities requiring copies of the SDS.

6.4 Rounding-off specified limits. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value should be rounded off to "the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29, Standard Recommended Practice for Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values.

6.5 Process control document. A process control document (manufacturing instructions) should be submitted for approval to the Technical Agency (see 6.6) prior to the first article test. The document should contain a description of the process, all materials used, process conditions/procedures, and production equipment used to produce zirconium carbide meeting the requirements of this specification. In addition, whenever there is a proposed change such as the source of materials, process conditions, procedures, etc., from those in the previous process control document, the Technical Agency (6.6) should be notified of the new proposed process

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and process control change(s) and the reason for the change(s) with not less than 15 business days for Government approval prior to implementation.

6.6 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC – AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.7 Subject term (key word) listing.

Igniter
Base bleed
Propellant
Inhibitor

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodian:
Army – AR

Preparing activity:
Army – AR
(Project 6810-2021-003)

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